

# Rahul Patnala

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## Professional Summary

AI/ML Engineer specializing in **Generative AI, Agentic AI, and LLM orchestration**. Experienced in designing multi-agent LangGraph workflows, Retrieval-Augmented Generation (RAG) pipelines, hybrid retrieval systems, LLM routing, guardrails, and tool-enabled AI applications.

## Education

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|-------------------------------------------------------------------------------------------------------------|--------------------|
| <b>B.Tech in Computer Science and Engineering (AI &amp; ML)</b> , GMR Institute of Technology<br>CGPA: 9.04 | <b>2022 – 2026</b> |
| <b>Intermediate</b> , Sri Viswa Junior College<br>Percentage: 96.8%                                         | <b>2020 – 2022</b> |
| <b>SSC</b> , Srujana Vidyalaya<br>CGPA: 9.82                                                                | <b>2019 – 2020</b> |

## Technical Skills

**Programming:** Python, Java, SQL, JavaScript, C, C++

**Generative AI:** Generative AI, Agentic AI, LLMs, LangGraph, LangChain, RAG, Prompt Engineering, Structured Outputs, LoRA Fine-Tuning, Model Context Protocol (MCP)

**Machine Learning:** Machine Learning, Deep Learning, NLP, Scikit-Learn, TensorFlow, NumPy, Pandas

**Backend:** Spring Boot, FastAPI, Flask, REST APIs, Microservices, React, Streamlit

**AI Infrastructure:** Qdrant, ChromaDB, PostgreSQL, Supabase, Hugging Face, BGE Embeddings, FlashRank

**Evaluation & Tools:** RAGAS, LangSmith, NVIDIA NeMo Guardrails, Portkey, Git, GitHub

## Experience

**GenC Intern – Cognizant** **Feb 2026 – Jun 2026**

- **ComNet360 Platform:** Developing an enterprise communications system using a Spring Boot microservices architecture, featuring an API Gateway, JWT-based RBAC, and automated SLA breach detection.
- Building a React-based frontend with role-specific dashboards and integrating multiple backend services through a centralized gateway for real-time incident analytics.

## Projects

**TourAlly – Multi-Agent AI Travel Planner** **GitHub**

- Built a multi-agent LangGraph workflow using a supervisor-specialist architecture to dynamically route requests across Flights, Hotels, Weather, and Budget agents powered by Llama models through Groq.
- Integrated real-time information using Model Context Protocol (MCP), AviationStack, Tavily, and OpenWeather APIs with dynamic currency conversion for itinerary cost normalization.
- Implemented input-validation guardrails, human-in-the-loop checkpoints, Supabase state persistence, and LangSmith observability for reliable multi-turn agent execution.

**Banking Knowledge Intelligence Platform (BKIP)** **GitHub**

- Architected a stateful Agentic RAG system using LangGraph and FastAPI for querying banking policies, RBI circulars, and internal guidelines with dynamic intent routing.
- Built a hybrid retrieval pipeline using Qdrant, BGE embeddings, and FlashRank ONNX reranking to improve context precision and retrieval quality.
- Added NVIDIA NeMo Guardrails, PII protection, LLM fallback routing via Portkey, and RAGAS/LangSmith evaluation for production-oriented reliability and observability.

## Achievements

- **1st Place – DataCraze, Stepcone 2025:** Analyzed a 100K+ Twitter dataset to identify actionable sentiment trends and developed an interactive analytics dashboard. **GitHub**
- **1st Place – DataViz, Stepcone 2024:** Developed a geographic sales analytics dashboard using feature engineering, custom mapping, and automated business summaries. **GitHub**

## Certifications

- **AI & Data Science Program** – DRISHTI CPS, IIT Indore (2025)
- **Deep Learning** – NPTEL IIT Ropar (2025), **Top 1% Performer**
- **Data Science and Machine Learning Bootcamp** – Udemy (2025)